# SOFTWARE REQUIREMENTS SPECIFICATION

## HealthAI Co-Creation & Innovation Platform

**Version 2.0**

Course: SENG 384 – Software Project IV

Institution: Çankaya University


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## Revision History

| Version | Date | Author(s) | Description |
|---|---|---|---|
| 1.0 | March 2026 | Senanur Çinkil | Initial SRS draft |
| 2.0 | April 21, 2026 | Emre Tek | Full sections added: 3.2–3.4, UC-01, UC-02, data model, RTM |

# 1. Introduction

## 1.1 Purpose

This Software Requirements Specification (SRS) document describes the complete functional and non-functional requirements of the HealthAI Co-Creation & Innovation Platform. The system is designed to enable structured collaboration between engineers and healthcare professionals by providing a platform for partner discovery and communication.

## 1.2 Scope

The platform allows users to create posts, discover potential collaborators, and initiate contact. The system does NOT store intellectual property files, patient data, or sensitive medical information. Meetings are conducted outside the platform using external tools such as Zoom or Microsoft Teams.

## 1.3 Definitions

| Term | Definition |
|---|---|
| SRS | Software Requirements Specification document |
| RBAC | Role-Based Access Control, restricting access based on user roles |
| GDPR | General Data Protection Regulation for data privacy and protection |
| NDA | Non-Disclosure Agreement used to protect confidential information |
| JWT | JSON Web Token used for secure session and authentication management |
| WCAG | Web Content Accessibility Guidelines for accessible web design |
| .edu | Institutional email domain required for user registration |

## 1.4 Intended Audience

This document is intended for the development team and the course instructor.

# 2. Overall Description

## 2.1 Product Perspective

Currently, collaboration between engineers and healthcare professionals depends heavily on personal networks, making it inefficient and limited. The platform provides a structured solution by enabling users to publish collaboration posts, discover relevant partners, and initiate communication in a controlled and systematic way.

## 2.2 User Roles

- Engineer – Creates posts and collaborates with healthcare professionals who express interest.
- Healthcare Professional – Creates or browses posts and interacts with engineers.
- Admin – Moderates content, manages users, and oversees platform activities.

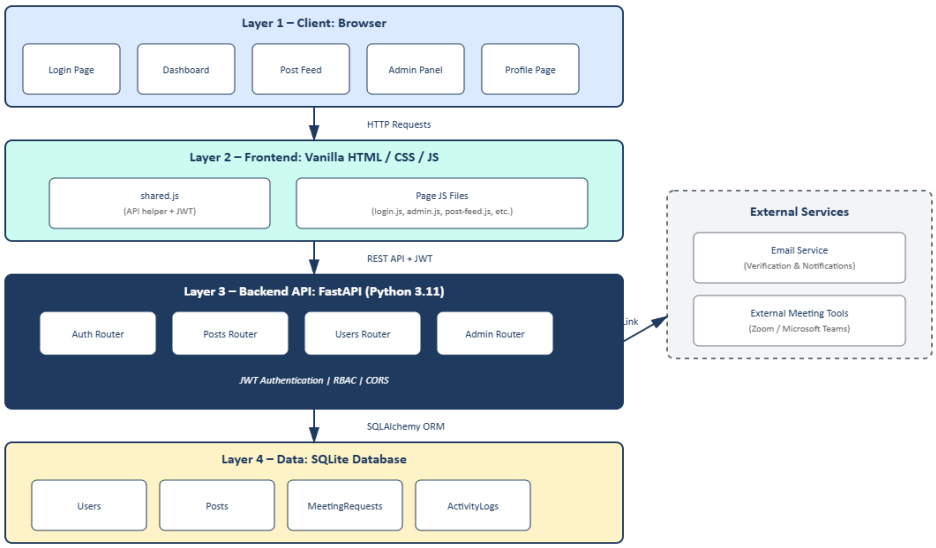
## 2.3 Assumptions

- Users must have a valid institutional (.edu) email address.
- Users have access to external meeting tools such as Zoom or Microsoft Teams.
- Users use modern web browsers (Chrome, Firefox, Edge).

## 2.4 Constraints

- The system does not support file uploads.
- The system does not store patient data.
- Meetings are conducted outside the platform.

**Figure 3 – System Architecture Diagram**



# 3. Functional Requirements

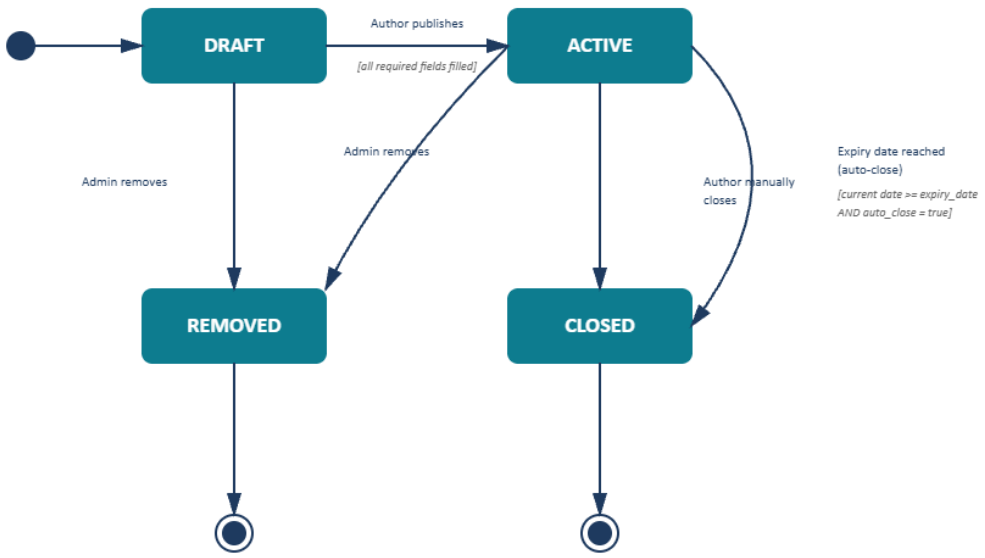
## 3.1 User Registration & Access Control

| FR-ID | Description | Priority | Source |
|---|---|---|---|
| FR-01 | Registration must be performed using an institutional (.edu) email address only. | High | Brief 4.1 |
| FR-02 | Email verification is required for account activation. | High | Brief 4.1 |
| FR-03 | Users must select a role during registration (Engineer or Healthcare Professional). | High | Brief 4.1 |
| FR-04 | Role-Based Access Control (RBAC) must be enforced, ensuring users can only perform actions permitted by their role. | High | Brief 4.1 |
| FR-05 | The Admin role cannot be selected during registration and must be assigned through a separate process. | Medium | Brief 4.1 |
| FR-06 | User sessions must be managed securely using JWT tokens. | High | Brief 5.2 |

## 3.2 Post Management

| FR-ID | Description | Priority | Source |
|---|---|---|---|
| FR-10 | Authenticated users (Engineer or Healthcare Professional) shall be able to create a collaboration post. | High | Brief 4.2 |
| FR-11 | Each post shall have a lifecycle with the following statuses: draft, active, closed. | High | Brief 4.2 |
| FR-12 | The post author shall be able to edit their own post while it is in draft or active status. | Medium | Brief 4.2 |
| FR-13 | The post author shall be able to manually close their post at any time. | Medium | Brief 4.2 |
| FR-14 | The system shall automatically expire (close) a post after a user-defined expiry date. | Medium | Brief 4.2 |
| FR-15 | A post shall include the following fields: title, domain, short explanation (≤ 500 chars), expertise needed, project stage, confidentiality level, city, country, expiry date. | High | Brief 4.2 |
| FR-16 | The post author shall be able to mark a post as confidential, which hides full details from unauthenticated users. | Low | Brief 4.2 |

Figure 4 – Post Lifecycle State Diagram




## 3.3 Post Discovery & Filtering

| FR-ID | Description | Priority | Source |
|---|---|---|---|
| FR-20 | Authenticated users shall be able to browse all active posts in a paginated feed. | High | Brief 4.3 |
| FR-21 | The post feed shall support filtering by: domain, project stage, city/country, and confidentiality level. | High | Brief 4.3 |
| FR-22 | The post feed shall support keyword search over title and explanation fields. | Medium | Brief 4.3 |
| FR-23 | Each post card in the feed shall display: title, domain, stage, city, and days remaining until expiry. | Medium | Brief 4.3 |
| FR-24 | Users shall be able to view the full detail of a post by navigating to a dedicated post detail page. | High | Brief 4.3 |

## 3.4 Meeting Request Flow

| FR-ID | Description | Priority | Source |
|---|---|---|---|
| FR-30 | Authenticated users shall be able to express interest in a post by submitting a meeting request. | High | Brief 4.4 |
| FR-31 | A meeting request shall include: proposed time slots (up to 3), a short message, and an NDA acceptance checkbox. | High | Brief 4.4 |
| FR-32 | The post author shall be able to view all meeting requests received for their post. | High | Brief 4.4 |
| FR-33 | The post author shall be able to update the meeting request | High | Brief 4.4 |

| | | | |
|---|---|---|---|
| | status to: pending, scheduled, or rejected. | | |
| FR-34 | When a meeting is scheduled, the system shall display the agreed meeting link on the post detail page. | Medium | Brief 4.4 |
| FR-35 | A user shall not be able to submit more than one meeting request per post. | Medium | Brief 4.4 |

### 3.5 Administrative Dashboard

| FR-ID | Description | Priority | Source |
|---|---|---|---|
| FR-40 | The admin shall be able to view all posts filtered by status, domain, and date. | High | Brief 4.5 |
| FR-41 | The admin shall be able to remove inappropriate posts. | High | Brief 4.5 |
| FR-42 | The admin shall be able to list users filtered by role. | Medium | Brief 4.5 |
| FR-43 | The admin shall be able to suspend user accounts. | High | Brief 4.5 |
| FR-44 | The admin shall be able to view system statistics including total posts, active users, and daily actions. | Medium | Brief 4.5 |

### 3.6 Activity Logging & Audit Trail

| FR-ID | Description | Priority | Source |
|---|---|---|---|
| FR-50 | The system shall log critical actions including login, post creation, post deletion, meeting request submission, and account deletion. | High | Brief 4.6 |
| FR-51 | Each log entry shall include timestamp, user ID, action type, and target resource ID. | High | Brief 4.6 |
| FR-52 | The admin shall be able to filter logs by date range and user. | Medium | Brief 4.6 |
| FR-53 | The admin shall be able to export logs in CSV format. | Medium | Brief 4.6 |

## 4. Non-Functional Requirements

| ID | Requirement | Metric | Category |
|---|---|---|---|
| NFR-01 | Page load time shall be less than 3 seconds. | < 3 seconds | Performance |
| NFR-02 | The system shall support at least 100 concurrent users. | ≥ 100 users | Performance |
| NFR-03 | API response time shall be less than 500 milliseconds. | < 500 ms | Performance |
| NFR-04 | All communication shall be secured using HTTPS protocol. | 100% HTTPS | Security |
| NFR-05 | User authentication shall be managed using JWT tokens. | Token expiry: 24 hrs | Security |
| NFR-06 | The system shall implement Role-Based Access Control (RBAC). | 3 roles supported | Security |
| NFR-07 | Passwords shall meet complexity requirements. | Min 8 chars, uppercase, number | Security |
| NFR-08 | Users shall be able to delete their account and all personal data. | ≤ 30 days deletion | GDPR / Privacy |
| NFR-09 | Users shall be able to export their data in JSON format. | JSON download available | GDPR / Privacy |
| NFR-10 | The system shall not store sensitive medical or patient data. | 0 sensitive data stored | GDPR / Privacy |
| NFR-11 | The system shall comply with WCAG 2.1 AA accessibility standards. | WCAG 2.1 AA | Accessibility |
| NFR-12 | The system shall support keyboard navigation. | Full keyboard support | Accessibility |
| NFR-13 | The interface shall be easy to understand for new users. | ≤ 20 sec comprehension | Usability |
| NFR-14 | Error messages shall be clear and user-friendly. | 100% readable errors | Usability |

# 5. Use Cases

## 5.1 UC-01: User Registers and Creates a Post

**Name:** UC-01: User Registers and Creates a Post

**Actor(s):** Engineer, Healthcare Professional

**Precondition:** The user has a valid .edu email address and has not yet registered.

**Main Flow:** 1. The user navigates to /register.html.
2. The user enters name, .edu email, password, role, institution, expertise, and city.
3. The system validates the email domain (must end in .edu).
4. The system sends a verification email to the provided address.
5. The user clicks the verification link.
6. The account is activated and the user is redirected to /login.html.
7. The user logs in and is redirected to /dashboard.html.
8. The user navigates to /create-post.html.
9. The user fills in all required post fields and submits.
10. The post is saved with status 'active' and appears in the feed.

**Postcondition:** A new post is published and visible in the post feed.

**Alternative Flow:** 3a. If the email domain is not .edu, the system displays an error and halts registration.
4a. If the verification email is not received, the user may request a resend.
9a. If required fields are missing, the system highlights the empty fields.

## 5.2 UC-02: User Discovers Posts and Requests a Meeting

**Name:** UC-02: User Discovers Posts and Requests a Meeting

**Actor(s):** Engineer, Healthcare Professional

**Precondition:** The user is logged in with an activated account.

**Main Flow:** 1. The user navigates to /post-feed.html.
2. The user applies filters (domain, stage, city) or searches by keyword.
3. The system returns matching posts in a paginated list.
4. The user clicks a post to view its full details on /post-detail.html.
5. The user clicks 'Express Interest'.
6. The user is redirected to /meeting-request.html.
7. The user fills in proposed time slots, a short message, and accepts the NDA checkbox.
8. The user submits the meeting request.
9. The system saves the request with status 'pending' and logs the action.

**Postcondition:** A meeting request is created and visible to the post author.

**Alternative Flow:** 5a. If the user has already submitted a request for this post, the button is disabled.
7a. If the NDA checkbox is not checked, the form cannot be submitted.

### 5.3 UC-03: Admin Manages System

**Name:** UC-03: Admin Manages System

**Actor(s):** Admin

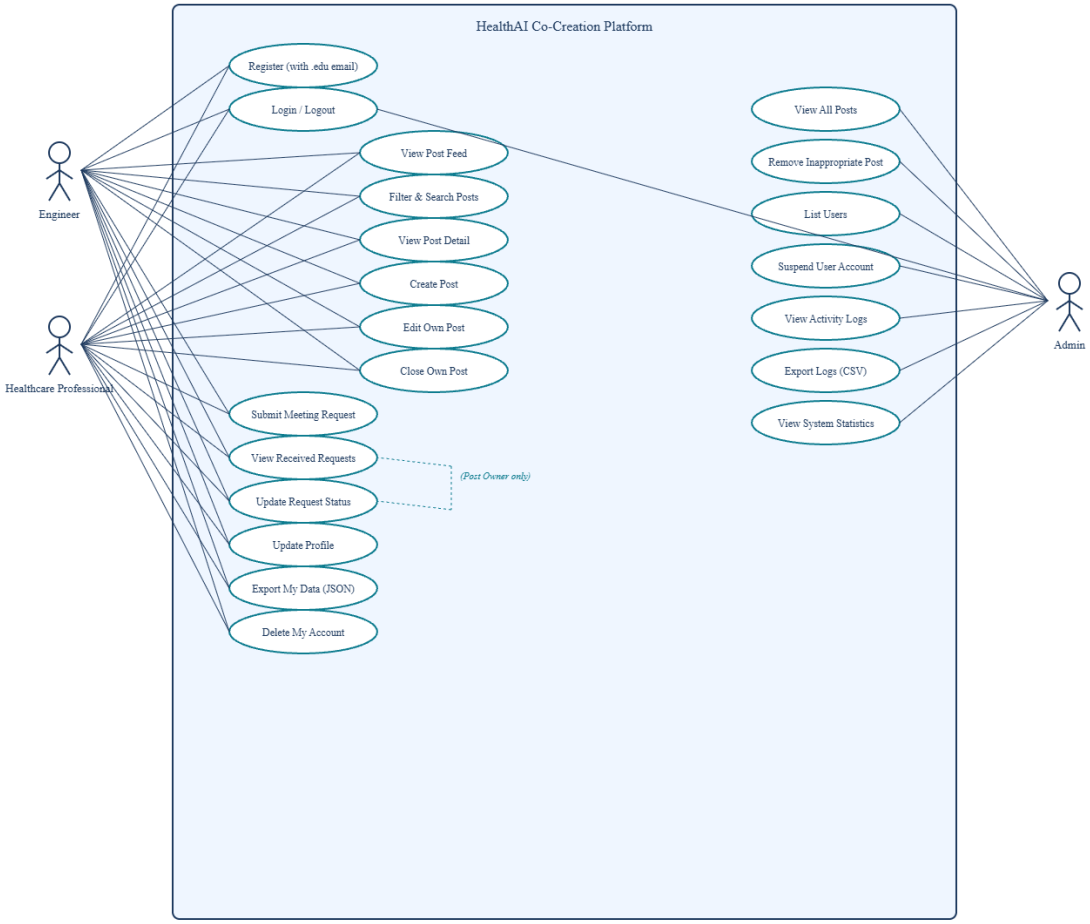
**Precondition:** The admin must be logged in.

**Main Flow:** 1. The admin opens the admin panel (/admin.html).
2. The admin views users and posts.
3. The admin applies filters to the data.
4. The admin removes inappropriate posts.
5. The admin suspends user accounts if necessary.
6. The admin reviews activity logs.
7. The admin exports logs as CSV.

**Postcondition:** System data is managed and updated.

**Alternative Flow:** 3a. If no data is found, the system displays an empty state message.

Figure 2 – Use Case Diagram



# 6. Data Model

## 6.1 Entity Descriptions

The system is based on four main entities: Users, Posts, MeetingRequests, and ActivityLogs. These entities represent the core structure of the platform and define how users interact with the system.

### Users

Stores all registered user information, including authentication details, role, and profile data. Each user can create posts, send meeting requests, and generate activity logs.

| Column | Type | Notes |
|---|---|---|
| id | INTEGER | Primary Key |
| name | VARCHAR(120) | Full name |
| email | VARCHAR(200) | Unique .edu email |
| hashed_password | TEXT | bcrypt hash |
| role | VARCHAR(30) | engineer \| doctor \| admin |
| institution | VARCHAR(200) | University/hospital |
| expertise | TEXT | Area of expertise |
| city | VARCHAR(100) | |
| is_suspended | BOOLEAN | Default: false |
| is_verified | BOOLEAN | Email-verified flag |
| created_at | DATETIME | Auto-set on insert |

### Posts

Represents collaboration opportunities created by users. Each post belongs to a single user and contains detailed information such as domain, required expertise, project stage, and confidentiality level.

| Column | Type | Notes |
|---|---|---|
| id | INTEGER | Primary Key |
| user_id | INTEGER | FK → Users.id |
| title | VARCHAR(200) | |
| domain | VARCHAR(100) | Medical domain |
| explanation | TEXT | ≤ 500 characters |
| expertise_needed | TEXT | |
| stage | VARCHAR(50) | idea \| prototype \| pilot \| scaling |
| confidentiality | VARCHAR(20) | public \| confidential |
| city | VARCHAR(100) | |
| country | VARCHAR(100) | |
| expiry_date | DATE | Auto-close date |
| auto_close | BOOLEAN | |
| status | VARCHAR(20) | draft \| active \| closed |
| created_at | DATETIME | |

### MeetingRequests

Represents meeting requests sent by users in response to posts. Each request is linked to both a user and a post, and includes proposed time slots, NDA acceptance, and request status.

| Column | Type | Notes |
|---|---|---|
| id | INTEGER | Primary Key |
| post_id | INTEGER | FK → Posts.id |
| requester_id | INTEGER | FK → Users.id |
| proposed_slots | TEXT | JSON array of ISO datetimes |
| message | TEXT | Short message |
| nda_accepted | BOOLEAN | |
| status | VARCHAR(20) | pending \| scheduled \| rejected |
| created_at | DATETIME | |

### ActivityLogs

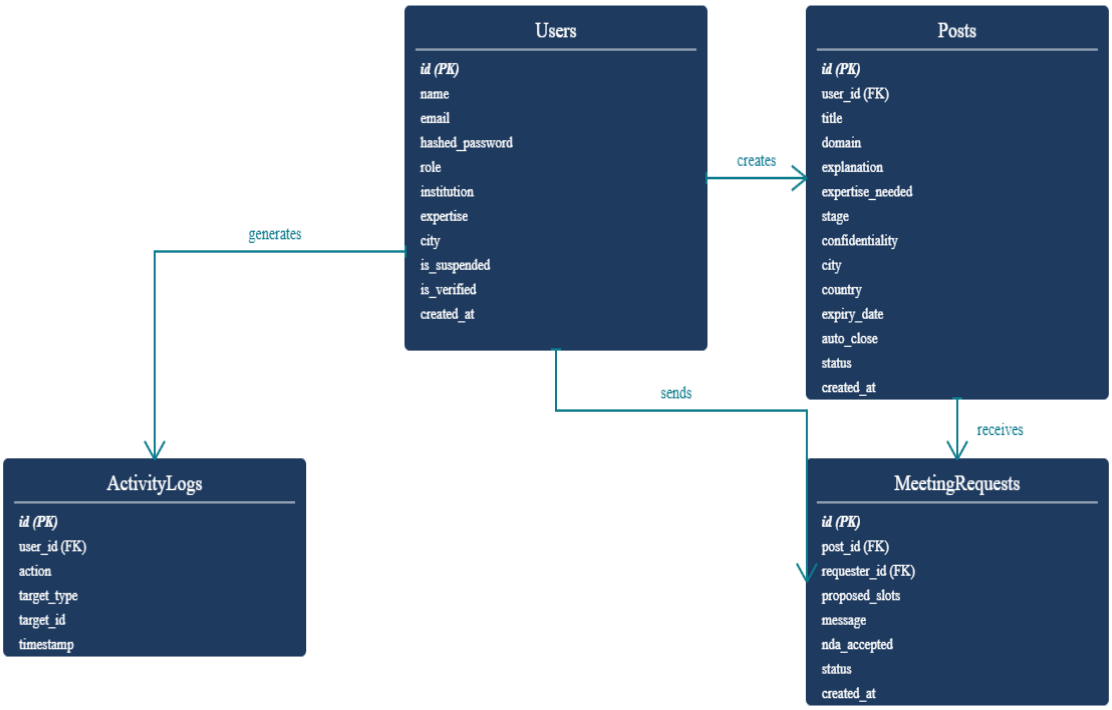
Stores system activity records. Each log entry tracks user actions such as login, post creation, meeting requests, and account-related operations.

| Column | Type | Notes |
|---|---|---|
| id | INTEGER | Primary Key |
| user_id | INTEGER | FK → Users.id |
| action | VARCHAR(100) | e.g. POST_CREATED |
| target_type | VARCHAR(50) | e.g. post \| user \| meeting |
| target_id | INTEGER | ID of affected resource |
| timestamp | DATETIME | Auto-set on insert |

## 6.2 Entity Relationships

- One User can create multiple Posts (1-to-many)
- One Post can have multiple MeetingRequests (1-to-many)
- One User can send multiple MeetingRequests (1-to-many)
- One User can generate multiple ActivityLogs (1-to-many)

Figure 1 – ER Diagram of the HealthAI Platform



# 7. Interface Requirements

## 7.1 User Interface (UI)

| Page | URL | Description |
|---|---|---|
| Register | /register.html | New user registration with .edu email validation |
| Login | /login.html | User login page |
| Dashboard | /dashboard.html | Main page showing posts and summary |
| Post Feed | /post-feed.html | List of posts with filtering |
| Create Post | /create-post.html | Post creation form |
| Post Detail | /post-detail.html | Detailed post view |
| Meeting Request | /meeting-request.html | Meeting request workflow |
| Profile | /profile.html | Profile management and GDPR actions |
| Admin Panel | /admin.html | Admin dashboard for managing users, posts, and logs |

## 7.2 External System Interfaces

- Email service is used for account verification and notifications.
- Meetings are conducted via external platforms such as Zoom or Microsoft Teams.
- The system only provides meeting links and does not host meetings.

## 7.3 API Interface

| Endpoint | Method | Description | Auth |
|---|---|---|---|
| /api/auth/register | POST | Register new user | None |
| /api/auth/login | POST | Login and receive JWT | None |
| /api/users/me | GET | Get current user profile | JWT |
| /api/users/me | PATCH | Update profile | JWT |
| /api/users/me | DELETE | Delete account and data | JWT |
| /api/users/me/export | GET | Export personal data as JSON | JWT |
| /api/posts | GET | List posts with filters | JWT |
| /api/posts | POST | Create a new post | JWT |
| /api/posts/{id} | PATCH | Update own post | JWT |
| /api/posts/{id} | DELETE | Delete own post | JWT |
| /api/posts/{id}/meeting-requests | POST | Submit meeting request | JWT |
| /api/admin/posts | GET | Admin: list all posts with filters | Admin |
| /api/admin/posts/{id} | DELETE | Admin: remove a post | Admin |
| /api/admin/users | GET | Admin: list users with role filter | Admin |
| /api/admin/users/{id}/suspend | PATCH | Admin: suspend user | Admin |
| /api/admin/logs | GET | Admin: view activity logs | Admin |
| /api/admin/stats | GET | Admin: system statistics | Admin |

## 8. Requirements Traceability Matrix

| Requirement ID | Description | Source | Use Case |
|---|---|---|---|
| FR-01 | .edu email requirement | Brief 4.1 | UC-01 |
| FR-02 | Email verification | Brief 4.1 | UC-01 |
| FR-03 | Role selection on register | Brief 4.1 | UC-01 |
| FR-04 | RBAC enforcement | Brief 4.1 | UC-01, UC-02, UC-03 |
| FR-05 | Admin role not selectable | Brief 4.1 | - |
| FR-06 | JWT session management | Brief 5.2 | UC-01, UC-02 |
| FR-10 | Create post | Brief 4.2 | UC-01 |
| FR-11 | Post lifecycle (draft→active→closed) | Brief 4.2 | UC-01 |
| FR-12 | Edit own post | Brief 4.2 | UC-01 |
| FR-13 | Manual close post | Brief 4.2 | UC-01 |
| FR-14 | Post expiry (auto-close) | Brief 4.2 | - |
| FR-15 | Post required fields | Brief 4.2 | UC-01 |
| FR-16 | Confidential post flag | Brief 4.2 | UC-01 |
| FR-20 | Post filtering / feed | Brief 4.3 | UC-02 |
| FR-21 | Filter by domain/stage/city | Brief 4.3 | UC-02 |
| FR-22 | Keyword search | Brief 4.3 | UC-02 |
| FR-23 | Post card content | Brief 4.3 | UC-02 |
| FR-24 | Post detail page | Brief 4.3 | UC-02 |
| FR-30 | Express interest | Brief 4.4 | UC-02 |
| FR-31 | Meeting request fields | Brief 4.4 | UC-02 |
| FR-32 | View received requests | Brief 4.4 | UC-02 |
| FR-33 | Meeting scheduled status | Brief 4.4 | UC-02 |
| FR-34 | Meeting link display | Brief 4.4 | UC-02 |
| FR-35 | One request per post per user | Brief 4.4 | UC-02 |
| FR-40 | Admin post management | Brief 4.5 | UC-03 |
| FR-41 | Remove post | Brief 4.5 | UC-03 |
| FR-42 | User listing | Brief 4.5 | UC-03 |
| FR-43 | Suspend user | Brief 4.5 | UC-03 |
| FR-44 | System statistics | Brief 4.5 | UC-03 |
| FR-50 | Activity logging | Brief 4.6 | - |
| FR-51 | Log details | Brief 4.6 | - |
| FR-52 | Log filtering | Brief 4.6 | UC-03 |
| FR-53 | Export logs CSV | Brief 4.6 | UC-03 |